Time-delayed Information Flow in Foreign Exchange Data

A.M.Y.R. SOUSA[1], H. TAKAYASU[2,3], and M. TAKAYASU[1]

[1]Department of Computational Intelligence and Systems Science, Interdisciplinary Graduate School of Science and Engineering, Tokyo Institute of Technology, G3-52 4259 Nagatuta-cho, Yokohama 226-8502, Japan

[2]Sony Computer Science Laboratories Inc., 3-14-13 Higashigotanda, Shinagawa-ku, Tokyo 141-0022, Japan

[3]Meiji Institute for Advanced Study of Mathematical Science, Meiji University, 4-21-1, Nakano, Nakano-ku, Tokyo, 164-8525, Japan

E-mail: [1]yamashita.a.ai@m.titech.ac.jp

Abstract

We propose a mutual information based method to access the correlation among sign time series of foreign exchange data for EUR, JPY and USD. Searching for correlations when a time shift is applied, we were able to detect the time-delay and directionality of information flow between two markets. Future works include the construction of a directed network for currency markets and the analysis of its time evolution.

Keyword: Foreign Exchange, Mutual Information, Information Flow

1 Introduction

The foreign exchange is a continuously open market with the largest transaction volume among the financial markets. Interdependence and correlation between price fluctuations in this market are topics of intense study. Here, we focus on exploring also the non-linear correlations using the mutual information as a measurement.

2 Material and Methods

We used the Electronic Broking Services data for currency pairs by ICAP and analyzed the average price of the markets EUR/JPY, EUR/USD and USD/JPY in a resolution of 100 milliseconds for one day period (2011/03/08-09).
We converted the average price differences series to the sign series by observing the sign of the change, positive or negative. Next we estimated the probabilities of positive and negative changes for each series as well as the joint probability. Finally we determined the mutual information and the normalized global coefficient, used as a measurement of correlation.

3 Results and Perspectives

We calculated the mutual information among the markets EUR/JPY, EUR/USD and USD/JPY with temporal shifts. We observed that there are cases in which the maximum correlation was not at shift 0, i.e., the past values of one market influence others. This shifted maximum correlation shows not only the interdependence but also the directionality of information flow.
Results suggest the possibility of building a directed network of currency markets, elucidating the influence structure. Moreover, we intend to analyze the time evolution of this network, verifying if there are changes in the information hubs or in the direction of information flow, providing a dynamic description of the correlations.

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